

Stream Editing with sed — Security Log Analysis

Part A — Authentication Log

Exercise 1: To print

Print the first entry, the fifth entry, lines 2 to 6, the last entry, every second entry from line 1, all Failed password entries, all Accepted password entries, and the range from the first failed login to the authentication failure.

```
sanjana@ubuntu-vm:~/sed_lab$ sed -n '1p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
sanjana@ubuntu-vm:~/sed_lab$ sed -n '5p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
sanjana@ubuntu-vm:~/sed_lab$ sed -n '2,6p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
sanjana@ubuntu-vm:~/sed_lab$ sed -n '$p' auth.log
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed -n '1~2p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/Failed password/,/authentication failure/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

The **-n** option is important here. Without it sed prints every line by default and **p** then prints the matched lines a second time, so the output has duplicates. **\$** is the shortcut for the last line, and **1~2** is the step form, which gives lines 1, 3, 5, 7 and 9.

The last command uses two patterns as a range, so printing starts at the first failed password (line 2) and stops at the sudo authentication failure (line 7).

Exercise 2: Filtering Security Events

Display failed logins, successful logins, root account entries, Invalid user entries, all events for 172.16.0.8, events without Accepted password, events except sudo, and Failed password or Invalid user using multiple -e.

```
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/root/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/Invalid user/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/172\.16\.0\.8/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/Accepted password/!p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/sudo/!p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
```

```
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed -n -e '/Failed password/p' -e '/Invalid user/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

The **!** after a pattern reverses it, so **/sudo/!p** prints every line that does not mention sudo. The dots in the IP are escaped as **\.** because a plain dot matches any character, so without the backslashes it would also match strings like 172x16y0z8. The last command uses two **-e** options to run two print commands in one pass.

Exercise 3: Removing Irrelevant Entries

Without modifying the original file, display the log after deleting: successful logins, Connection closed entries, the first three entries, lines 4 to 7, entries for 192.168.1.15, every second entry from line 1, and all events that do not contain Failed password.

```
sanjana@ubuntu-vm:~/sed_lab$ sed '/Accepted password/d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed '/Connection closed/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
sanjana@ubuntu-vm:~/sed_lab$ sed '1,3d' auth.log
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed '4,7d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
```

```

Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed '/192\.168\.1\.\.15/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed '1~2d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed '/Failed password/!d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

```

None of these commands use **-i**, so the original **auth.log** is never touched. sed only prints the edited version on the screen. The last command deletes everything that is not a failed password, so only those lines are kept.

Exercise 4: Classifying Security Events

Replace Accepted password, Failed password, Invalid user and authentication failure with labels; replace root with PRIVILEGED_USER; replace every occurrence of server with authentication-server.

```
sanjana@ubuntu-vm:~/sed_lab$ sed 's/Accepted password/[SUCCESSFUL LOGIN]/' auth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed 's/Failed password/[FAILED LOGIN]/' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed 's/Invalid user/[INVALID USER]/' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: [INVALID USER] hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
```

```

Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed 's/authentication failure/[AUTHENTICATION FAILURE]/' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: [AUTHENTICATION FAILURE] for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed 's/root/PRIVILEGED_USER/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for PRIVILEGED_USER from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for PRIVILEGED_USER from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for PRIVILEGED_USER from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed 's/server/authentication-server/g' auth.log
Sep 10 09:10:12 authentication-server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 authentication-server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 authentication-server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 authentication-server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 authentication-server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 authentication-server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 authentication-server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 authentication-server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 authentication-server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 authentication-server sshd[1240]: Connection closed by 10.0.0.25

```

The **s** command works as **s/what to find/what to replace it with/**. The **g** flag was added on the last two commands so that every occurrence of root and server on a line is replaced, not just the first one. Without **g**, sed changes only the first match on each line.

Exercise 5: Masking Sensitive Information

Anonymise the log: mask 10.0.0.25 and 172.16.0.8, rename users admin and student, replace all digits with X, replace periods in IP addresses with hyphens, and replace all IP addresses with [REDACTED_IP].

```
sanjana@ubuntu-vm:~/sed_lab$ sed 's/10\.0\.0\.25/X.X.X.X/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from X.X.X.X
Sep 10 09:12:20 server sshd[1206]: Failed password for root from X.X.X.X
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by X.X.X.X
sanjana@ubuntu-vm:~/sed_lab$ sed 's/172\.16\.0\.8/SUSPICIOUS_IP/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from SUSPICIOUS_IP
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from SUSPICIOUS_IP
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed 's/admin/USER_A/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for USER_A from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for USER_A from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
```

```

Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed 's/student/USER_B/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for USER_B from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user USER_B
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed 's/[0-9]/X/g' auth.log
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for admin from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for student from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Invalid user hacker from XXX.XX.X.X
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for admin from XXX.XX.X.X
Sep XX XX:XX:XX server sudo[XXXX]: authentication failure for user student
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for analyst from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Connection closed by XX.X.X.XX
sanjana@ubuntu-vm:~/sed_lab$ sed -E 's/([0-9]{1,3})\.([0-9]{1,3})\.([0-9]{1,3})\.([0-9]{1,3})/\1-\2-\3-\4/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192-168-1-10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10-0-0-25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10-0-0-25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192-168-1-15

```

```

Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172-16-0-8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172-16-0-8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192-168-1-20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10-0-0-30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10-0-0-25
sanjana@ubuntu-vm:~/sed_lab$ sed -E 's/([0-9]{1,3})\.([0-9]{1,3})\.([0-9]{1,3})\.([0-9]{1,3})/[REDACTED_IP]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: Failed password for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]

```

[0-9] matches any single digit, so with **g** it replaces every number in the file, including the timestamps and process IDs. For the hyphen version **-E** turns on extended regex, so **()** and **{1,3}** can be used without backslashes. The four groups are captured and reused as **\1 \2 \3 \4**, so each part of the IP is kept and only the dots become hyphens.

Part B — Web-Server Log

Exercise 6: Printing Web Requests

Print requests with status 200, 401, 403, requests to /admin/login, POST requests, requests containing ../, <script>, OR, all requests except successful ones, and all requests from 10.0.0.25.

```
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/200/p' access.log
192.168.1.10 - GET /index.html 200
192.168.1.15 - GET /products.html 200
192.168.1.20 - GET /contact.html 200
192.168.1.25 - GET /about.html 200
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/401/p' access.log
10.0.0.25 - GET /admin/login 401
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/403/p' access.log
172.16.0.8 - GET /search?id=1'OR'1'='1 403
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.8 - GET /script?input=<script> 403
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/|/admin/login|p' access.log
10.0.0.25 - GET /admin/login 401
10.0.0.25 - POST /admin/login 401
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/POST/p' access.log
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/\.\.\./p' access.log
10.0.0.30 - GET /../../etc/passwd 403
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/<script>/p' access.log
172.16.0.8 - GET /script?input=<script> 403
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/OR/p' access.log
172.16.0.8 - GET /search?id=1'OR'1'='1 403
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/200!/p' access.log
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401

10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/^10\.0\.0\.25/p' access.log
10.0.0.25 - GET /admin/login 401
10.0.0.25 - POST /admin/login 401
```

The status codes are searched directly, for example /200/, and ! reverses the match for the “except successful” request. For /admin/login the delimiter is changed to a pipe with \|, otherwise every slash in the path would need escaping. The ^ anchors the last command to the start of the line so that only the source IP is matched.

Exercise 7: Labelling Suspicious Web Requests

Replace status 200 with [ALLOWED], 401 with [UNAUTHORIZED], 403 with [BLOCKED], /admin/login with /RESTRICTED_RESOURCE, GET with READ_REQUEST and POST with SUBMIT_REQUEST.

```
sanjana@ubuntu-vm:~/sed_lab$ sed 's/200/[ALLOWED]/' access.log
192.168.1.10 - GET /index.html [ALLOWED]
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html [ALLOWED]
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html [ALLOWED]
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html [ALLOWED]
sanjana@ubuntu-vm:~/sed_lab$ sed 's/401/[UNAUTHORIZED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login [UNAUTHORIZED]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
sanjana@ubuntu-vm:~/sed_lab$ sed 's/403/[BLOCKED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 [BLOCKED]
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd [BLOCKED]
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
```

```

10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> [BLOCKED]
192.168.1.25 - GET /about.html 200
sanjana@ubuntu-vm:~/sed_lab$ sed 's|/admin/login|/RESTRICTED_RESOURCE|' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
sanjana@ubuntu-vm:~/sed_lab$ sed 's/GET/READ_REQUEST/' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST ../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200
sanjana@ubuntu-vm:~/sed_lab$ sed 's/POST/SUBMIT_REQUEST/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200

```

```

10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - SUBMIT_REQUEST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - SUBMIT_REQUEST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200

```

Each status code is replaced directly with its label. For the URL substitution the separator is changed from / to |, which is much cleaner than escaping the slashes in the path.

Exercise 8: Grouping Security Operations

Use { } to run multiple commands on 401 entries, 403 entries and /admin/login requests.

```
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/401/{s/401/[UNAUTHORIZED]/;p}' access.log
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.9 - POST /login [UNAUTHORIZED]
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/403/{s/403/[BLOCKED]/;s/^[0-9.]\+/[REDACTED_IP]/;p}' access.log
[REDACTED_IP] - GET /search?id=1'OR'1'=1 [BLOCKED]
[REDACTED_IP] - GET ../../etc/passwd [BLOCKED]
[REDACTED_IP] - GET /script?input=<script> [BLOCKED]
sanjana@ubuntu-vm:~/sed_lab$ sed -n '|\admin/login|{s|\admin/login|/PROTECTED_PAGE|;s/^[0-9.]\+/[REDACTED_IP]/;p}'
cess.log
[REDACTED_IP] - GET /PROTECTED_PAGE 401
[REDACTED_IP] - POST /PROTECTED_PAGE 401
```

The curly braces let several commands run on the same matched line instead of making a separate pass for each one. The commands inside are separated by semicolons and run in order, so the substitution happens first and then **p** prints the already changed line. **^[0-9.]\+** means one or more digits or dots at the start of the line, which is the source IP field.

Exercise 9: Adding Incident Labels

Insert a heading before the first line of auth.log and append a footer after the last entry. Insert [WARNING] before failed logins, append Review this event after invalid users, insert [SUSPICIOUS WEB REQUEST] before 403 entries, append [REQUIRES INVESTIGATION] after 401 entries, and replace Connection closed entries and successful web requests with labels.

```
sanjana@ubuntu-vm:~/sed_lab$ sed '1i SECURITY AUTHENTICATION LOG' auth.log
SECURITY AUTHENTICATION LOG
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed '$a END OF SECURITY LOG' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
END OF SECURITY LOG
sanjana@ubuntu-vm:~/sed_lab$ sed '/Failed password/i [WARNING]' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
[WARNING]
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
[WARNING]
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
```

```
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
[WARNING]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
[WARNING]
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed '/Invalid user/a Review this event' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Review this event
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed '/403/i [SUSPICIOUS WEB REQUEST]' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
[SUSPICIOUS WEB REQUEST]
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
[SUSPICIOUS WEB REQUEST]
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
```

```

192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
[SUSPICIOUS WEB REQUEST]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
sanjana@ubuntu-vm:~/sed_lab$ sed '/401/a [REQUIRES INVESTIGATION]' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
[REQUIRES INVESTIGATION]
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
[REQUIRES INVESTIGATION]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
[REQUIRES INVESTIGATION]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
sanjana@ubuntu-vm:~/sed_lab$ sed '/Connection closed/c [CONNECTION TERMINATED]' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
[CONNECTION TERMINATED]

```

```

sanjana@ubuntu-vm:~/sed_lab$ sed '/200/c [NORMAL WEB REQUEST]' access.log
[NORMAL WEB REQUEST]
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
[NORMAL WEB REQUEST]
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
[NORMAL WEB REQUEST]
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
[NORMAL WEB REQUEST]

```

There are three different commands here. **i** puts a new line before the match, **a** puts one after it, and **c** throws the matched line away and replaces it with the new text. So the last two commands do not just label the entries, they wipe out the original content of those lines, which is why the

question says “replace” for them.

Exercise 10: Separating Log Entries with G

Add a blank line after every authentication event, after every failed login, after every blocked web request, and after every second entry starting with the first.

```
sanjana@ubuntu-vm:~/sed_lab$ sed 'G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10

Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25

Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15

Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

Sep 10 09:16:11 server sudo[1225]: authentication failure for user student

Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20

Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed '/Failed password/G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25

Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
```

```
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed '/403/G' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403

192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403

172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403

192.168.1.25 - GET /about.html 200
sanjana@ubuntu-vm:~/sed_lab$ sed '1~2G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10

Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
```

```
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

G appends the hold space to the end of the current line. The hold space is a second buffer that sed keeps aside, and since nothing was stored in it, it is empty, so appending it only adds a newline. That shows up as a blank line after the entry, and only for the lines that match the address given.

Exercise 11: Writing Events to Separate Files

Write failed logins, successful logins, invalid-user events, 401 and 403 web events, and all events related to 172.16.0.8 to separate files.

```
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/Failed password/w failed_logins.txt' auth.log
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/Accepted password/w successful_logins.txt' auth.log
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/Invalid user/w invalid_users.txt' auth.log
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/401/w unauthorized_requests.txt' access.log
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/403/w blocked_requests.txt' access.log
sanjana@ubuntu-vm:~/sed_lab$ sed -n '/172\.\16\.\0\.\8/w suspicious_ip.txt' auth.log access.log
sanjana@ubuntu-vm:~/sed_lab$ head -v failed_logins.txt successful_logins.txt invalid_users.txt
==> failed_logins.txt <==
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

==> successful_logins.txt <==
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20

==> invalid_users.txt <==
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
sanjana@ubuntu-vm:~/sed_lab$ head -v unauthorized_requests.txt blocked_requests.txt suspicious_ip.txt
==> unauthorized_requests.txt <==
10.0.0.25 - GET /admin/login 401
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401

==> blocked_requests.txt <==
172.16.0.8 - GET /search?id=1'OR'1'='1 403
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.8 - GET /script?input=<script> 403

==> suspicious_ip.txt <==
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
172.16.0.8 - GET /search?id=1'OR'1'='1 403
172.16.0.8 - GET /script?input=<script> 403
```

The **w** command writes the matching lines into a file instead of printing them. **head -v** shows the filename above each file so it is clear where one file ends and the next begins. **suspicious_ip.txt** contains both SSH login lines and web request lines because both log files were passed to the same sed command, which is useful when tracking one attacker across different services.

Exercise 12: Combining Operations

Label successful and failed logins; label 401 and 403 requests; delete successful logins and print failed logins; replace root with [PRIVILEGED_USER] and mask all IPs; replace GET and POST.

```
sanjana@ubuntu-vm:~/sed_lab$ sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/' -e 's/Failed password/[FAILED LOGIN]/' auth.log
uth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
sanjana@ubuntu-vm:~/sed_lab$ sed -e 's/401/[UNAUTHORIZED]/' -e 's/403/[BLOCKED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /search?id=1'OR'1'='1 [BLOCKED]
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd [BLOCKED]
172.16.0.9 - POST /login [UNAUTHORIZED]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /script?input=<script> [BLOCKED]
192.168.1.25 - GET /about.html 200
sanjana@ubuntu-vm:~/sed_lab$ sed -e '/Accepted password/d' -e '/Failed password/!d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
sanjana@ubuntu-vm:~/sed_lab$ sed -E -e 's/root/[PRIVILEGED_USER]/g' -e 's/([0-9]{1,3}\.){3}[0-9]{1,3}/[REDACTED_IP]/' auth.log
```

```
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
sanjana@ubuntu-vm:~/sed_lab$ sed -e 's/GET/READ_REQUEST/' -e 's/POST/SUBMIT_REQUEST/' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST ../../etc/passwd 403
172.16.0.9 - SUBMIT_REQUEST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - SUBMIT_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200
```

Using several **-e** options lets sed apply all the commands in one pass over the file, which is faster than piping one sed into another. The order matters, because the commands run top to bottom on every line, so if an earlier substitution changes text that a later one looks for, the later one will not find it.

Exercise 13: Using a sed Script File

Create `security_report.sed` with the required commands, execute it with `sed -f`, and save the output to `incident_report.txt`.

```
sanjana@ubuntu-vm:~/sed_lab$ cat > security_report.sed << 'EOF'
> 1i SECURITY INCIDENT REPORT
> s/Failed password/[FAILED LOGIN]/
> s/Invalid user/[INVALID USER ATTEMPT]/
> s/Accepted password/[SUCCESSFUL LOGIN]/
> s/[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}/[REDACTED_IP]/g
> $a END OF INCIDENT REPORT
> EOF
sanjana@ubuntu-vm:~/sed_lab$ sed -f security_report.sed auth.log
SECURITY INCIDENT REPORT
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: [INVALID USER ATTEMPT] hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
END OF INCIDENT REPORT
sanjana@ubuntu-vm:~/sed_lab$ sed -f security_report.sed auth.log > incident_report.txt
```

All the commands are kept in a `.sed` file, one per line, and `sed` runs them with the `-f` option. No quotes are needed because the shell never sees these commands. Inside a script file the IP pattern has to be written in basic regex with escaped braces `\{1,3\}`. The redirect `>` saves the same output, including the inserted heading and footer, into **incident_report.txt**.

Exercise 14: Modifying a Security Configuration File

Create `security.conf`, back it up, and use `sed -i` to harden the settings. Then use the safer `sed -i.bak` form.

```
sanjana@ubuntu-vm:~/sed_lab$ cat > security.conf << 'EOF'
> PermitRootLogin yes
> PasswordAuthentication yes
> MaxAuthTries 6
> FirewallStatus disabled
> LogLevel INFO
> GuestLogin enabled
> EOF
sanjana@ubuntu-vm:~/sed_lab$ cp security.conf security.conf.backup
sanjana@ubuntu-vm:~/sed_lab$ sed -i 's/PermitRootLogin yes/PermitRootLogin no/' security.conf
sanjana@ubuntu-vm:~/sed_lab$ sed -i 's/MaxAuthTries 6/MaxAuthTries 3/' security.conf
sanjana@ubuntu-vm:~/sed_lab$ sed -i 's/FirewallStatus disabled/FirewallStatus enabled/' security.conf
sanjana@ubuntu-vm:~/sed_lab$ sed -i 's/LogLevel INFO/LogLevel VERBOSE/' security.conf
sanjana@ubuntu-vm:~/sed_lab$ sed -i 's/GuestLogin enabled/GuestLogin disabled/' security.conf
sanjana@ubuntu-vm:~/sed_lab$ sed -i.bak 's/PasswordAuthentication yes/PasswordAuthentication no/' security.conf
sanjana@ubuntu-vm:~/sed_lab$ cat security.conf
PermitRootLogin no
PasswordAuthentication no
MaxAuthTries 3
FirewallStatus enabled
LogLevel VERBOSE
GuestLogin disabled
```

This is the only exercise where the file is really modified, using `-i`, so a backup was taken first with `cp`. Each command changes one setting. The safer form `-i.bak` does the same in-place edit but automatically keeps the previous version as `security.conf.bak`, without having to remember to run `cp` first. This only edits a sample file and does not configure the actual system.

Exercise 15: security_log_analyser.sh

Write a Bash script that takes a log filename, validates it, extracts failed and invalid-user attempts, masks IPs, adds the heading, [ALERT] labels and footer, saves the result in security_alert_report.txt and displays it.

```
sanjana@ubuntu-vm:~/sed_lab$ cat > security_log_analyser.sh << 'EOF'
> #!/bin/bash
>
> if [ $# -eq 0 ]; then
>     echo "Error: No log file provided."
>     echo "Usage: $0 <logfile>"
>     exit 1
> fi
>
> logfile="$1"
>
> if [ ! -e "$logfile" ]; then
>     echo "Error: File '$logfile' does not exist."
>     exit 1
> fi
>
> if [ ! -r "$logfile" ]; then
>     echo "Error: File '$logfile' is not readable."
>     exit 1
> fi
>
> output="security_alert_report.txt"
>
> echo "SECURITY ALERT REPORT" > "$output"
>
> sed -n -e '/Failed password/p' -e '/Invalid user/p' "$logfile" \
> | sed -E -e 's/([0-9]{1,3}\.){3}[0-9]{1,3}/[REDACTED_IP]/g' \
>     -e 's/^/[ALERT] /' >> "$output"
>
> echo "END OF REPORT" >> "$output"
```

```

>
> echo "Report saved to $output"
> echo
> cat "$output"
> EOF
sanjana@ubuntu-vm:~/sed_lab$ chmod +x security_log_analyser.sh
sanjana@ubuntu-vm:~/sed_lab$ ./security_log_analyser.sh
Error: No log file provided.
Usage: ./security_log_analyser.sh <logfile>
sanjana@ubuntu-vm:~/sed_lab$ ./security_log_analyser.sh nofile.log
Error: File 'nofile.log' does not exist.
sanjana@ubuntu-vm:~/sed_lab$ ./security_log_analyser.sh auth.log
Report saved to security_alert_report.txt

SECURITY ALERT REPORT
[ALERT] Sep 10 09:12:15 server sshd[1205]: Failed password for root from [REDACTED_IP]
[ALERT] Sep 10 09:12:20 server sshd[1206]: Failed password for root from [REDACTED_IP]
[ALERT] Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
[ALERT] Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
[ALERT] Sep 10 09:18:05 server sshd[1235]: Failed password for root from [REDACTED_IP]
END OF REPORT

```

With no argument \$# is 0, so the script prints the usage message and exits before using an argument that does not exist. A file that is missing fails the **-e** check, and the script stops before any sed command runs. For a valid file, the first sed keeps only the failed-password and invalid-user lines, and the second sed masks every IP and adds **[ALERT]** at the start of each line using **s/^/[ALERT] /**. The heading and footer come from **echo**, with **>** creating the file and **>>** adding to it.